

Republic of the Philippines
NATIONAL AVIATION ACADEMY OF THE PHILIPPINES
The National Professional Institution for Aviation
INSTITUTE OF COMPUTER STUDIES

July 27, 2026

Dear Respondent,

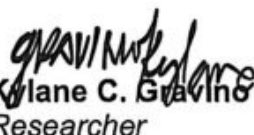
Greetings!

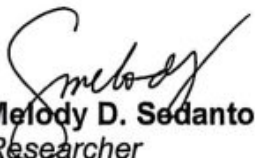
We are fourth-year Bachelor of Science in Aviation Information Technology (BSAIT) Section 2 students from the National Aviation Academy of the Philippines, conducting our capstone project entitled: **"A Web-Based Service Management System with Data Analytics Using Machine Learning Algorithms for Shoelotskey Villamor-Pasay"**. As part of our study, we respectfully invite you to evaluate the developed system. Your responses will help us assess the system's perceived usefulness, perceived ease of use, behavioral intention to use, and actual system use. Your feedback will provide valuable information for improving the system and determining its overall acceptability.

Your participation is entirely voluntary. You may choose not to answer any question or withdraw from the evaluation at any time without any penalty. All information you provide will be kept confidential and will be used solely for academic and research purposes.

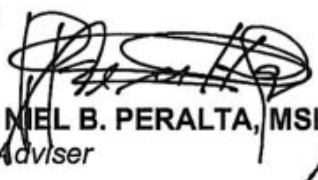
Respectfully,


Charmaine Rose G. Angeles
Researcher


Kyle C. Graving
Researcher


Melody D. Sedanto
Researcher

Noted by:


RICHARD MEL B. PERALTA, MSIT
Capstone Adviser


ASSOC. PROF. MARY ANN ABALLIAR-VISTA, MEAM
Academic Adviser

In accordance with the Data Privacy Act of 2012 (Republic Act No. 10173), all information collected through this questionnaire will be treated with the utmost confidentiality and will be used solely for academic and research purposes. Your identity will remain anonymous, and the information you provide will not be disclosed to any unauthorized person or organization. The collected data will be securely stored and disposed of after the study's completion.

By completing this questionnaire, you indicate your voluntary consent to participate in this study.

Consent Declaration

☒ I have read and understood the Data Privacy and Confidentiality Statement.

☒ I voluntarily agree to participate in this research study.

Thank you for your valuable time and participation.



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Capstone Title: "A Web-Based Service Management System with Data Analytics Using Machine Learning Algorithms for Shoelotskey Villamor-Pasay"

Objectives of the Study

This capstone project aims to design and develop the Web-Based Service Management System with Data Analytics using Machine Learning Algorithms in order to improve operational efficiency, data accuracy, service monitoring, and managerial decision-making.

Specific Objectives

Specifically, the project aims to achieve the following:

1. Develop and implement a centralized digital job order entry and management system that captures customer information, service specifications, and transaction records, thereby reducing average service record retrieval time from 15-30 minutes to less than 5 minutes.
2. Implement and evaluate a machine learning model for predicting service completion dates with a target accuracy of 85% or higher compared to manual estimation methods.
3. Generate a real-time analytics dashboard that generates automated daily, weekly, monthly, quarterly, and annual reports, reducing manual report preparation time from 4-6 hours monthly to less than 1 hour.
4. Evaluate the effectiveness of the developed system based on the ISO 25010 software quality standards, specifically in terms of:
 - 4.1 Functional suitability;
 - 4.2 Performance efficiency;
 - 4.3 Compatibility;
 - 4.4 Interaction Capability;
 - 4.5 Reliability;
 - 4.6 Security;
 - 4.7 Maintainability;



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4.8 Flexibility; and

4.9 Safety;

Additionally, the study will assess data accuracy and prediction reliability of the machine learning component.

5. Measure the level of user acceptance of the developed system among the staff and management of Shoelotskey using the Technology Acceptance Model (TAM), through a survey questionnaire measuring perceived usefulness, perceived ease of use, behavioral intention to use, and actual system use.
6. Recommend possible system enhancements by identifying improvements and additional features that may be implemented by future researchers to further enhance system functionality, data analytics capabilities, and user experience.



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Survey Questionnaire

Part I. Respondent Profile

Instructions: Please provide the following information by filling in the data or putting a (✓) next to your response.

Name: (Optional) YANG

Position at Shoelotskey: ☐ Owner ☒ Staff

Part II. Technology Acceptance Model (TAM) Evaluation

Instructions: Please read each statement carefully. Put a (✓) in the box that best shows how much you agree with each statement based on your experience using the Web-Based Service Management System for Shoelotskey Villamor-Pasay"

Scale	Interpretation	Description
4	Highly Acceptable	The system fully meets the evaluation criterion and performs exceptionally well.
3	Acceptable	The system meets the evaluation criterion and performs satisfactorily.
2	Less Acceptable	The system partially meets the evaluation criterion and requires improvement.
1	Not Acceptable	The system does not meet the evaluation criterion and requires significant improvement.

CRITERIA	4	3	2	1
Perceived Usefulness (PU) - This section evaluates whether the system helps users perform their tasks more effectively and efficiently.				
1. The system helps complete work more quickly.	✓			
2. The system makes it easier to manage customer records and job orders.	✓			
3. The system helps track services and inventory more quickly.	✓			
4. The system provides useful information for monitoring sales and shop performance.	✓			
5. The system reduces the need for paperwork and manual record-keeping.	✓			



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CRITERIA	4	3	2	1
Perceived Ease of Use (PEOU)- This section evaluates whether the system is easy to learn and use.				
1. The menus, buttons, and labels are easy to understand.	/			
2. The system is easy to learn.	/			
3. Adding and updating records in the system is easy.	/			
4. Information in the system is easy to find.	/			
5. The system is easy to use.	/			
Behavioral Intention to Use (BI)- This section evaluates the user's willingness and intention to continue using the system.				
1. The system is worth using for daily work.	/			
2. The system should be used as the main system in the shop.	/			
3. The system is worth recommending to other staff.	/			
4. The system is better than the previous way of managing shop work.	/			
5. The system should continue to be used at Shoelotskey.	/			
Actual System Use (ASU)- This section evaluates how often and how consistently the user uses the system in daily work.				
1. The system is used during work tasks.	/			
2. The system is used to record and manage customer job orders.	/			
3. The system is used to check inventory, service status, and reports.	/			
4. Work is completed using the system instead of paper records.	/			
5. The system is used regularly in shop operations.	/			



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PART III. Comments and Suggestions

Instructions: Please answer the questions below based on your experience using the system.

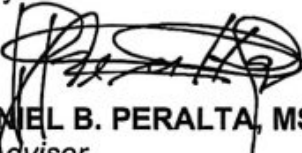
1. What feature of the system do you like the most?

2. Did you experience any problems while using the system? If yes, please describe them.

3. What improvements or new features would you like to see in the future?

"We assure you that your responses will be treated with the utmost confidentiality and used strictly for academic purposes."

Approved by:


RICHARD NIEL B. PERALTA, MSIT
Capstone Adviser


Signature of the Respondent

Date: 08/31/24

